

**SINGLE SOURCE DETERMINATION USING
SIMPLIFIED ACQUISITION PROCEDURES (SAP)
FOR AN ACTION NOT EXCEEDING THE SIMPLIFIED ACQUISITION THRESHOLD
(AUTHORITY: FAR 6.103-1)
Purchase Request Number: 143041-26-0005**

1. Agency and contracting activity.

Western Acquisition Division, Seattle
DOC/NOAA/AGO
7600 Sand Point Way
Seattle, WA 98115

2. Description of supplies or services required to meet agency needs (including the estimated value). Government-furnished passive integrated transponder (PIT) tags will be loaded into a single-use, specialized (proprietary) assemblage that consists of a needle and its base which connects to an injector gun. A small tray of 100 assemblages is needed to minimize the storage space needed for 30,000 tags while presenting a usable sized platform. These needle-tag assemblage must be compatible with a hand-held injector gun that will be used to tag the fish and includes a quick release mechanism. The PIT tags will be provided from another federal agency.

3. Identification of the single source or the brand name to be solicited.

Biomark, LLC,
705 S. 8th Street,
Boise ID 83702.

4. Supporting rationale. Only one source or brand name is reasonably available as detailed below:

- Compatibility to existing systems or equipment - For the current PIT-tagging process (half covered by this contract with the Corps of Engineers and half by a concurrent and already-funded Bonneville Power Administration (BPA) project), we use trays of 100 needle-tag assemblages purchased from Biomark; each assemblage is quickly extracted from the tray by an injector gun – also purchased from Biomark - which has a plunger that forces the tag into the fish after needle insertion. After tagging the fish, length and species data are taken on the fish, and the needle is then promptly ejected into a sharps container as disease protocols require single-use of needles. The fish tagger immediately grabs another assemblage with the injector gun and a fish to minimize the time fish are anesthetized while being tagged. The 16 Biomark injector guns we already own can only connect to the Biomark assemblage design. For this project, we are required to use the government-furnished PIT tags and not any other FDX ISO-B tags.
- Process –The fish used for both the BPA and Corps funded research projects are indiscriminately sent together into the tagging building and then processed simultaneously. Groups of fish for both studies are anesthetized so that the fish can be tagged. Fish taggers are rotated so that fish

from each study receive equal exposure to different taggers (this is a variable that we must control for as it can influence fish survival). Therefore, in order to avoid mistakes as fish are tagged for both projects simultaneously, fish taggers must use the same tag assemblages and associated injector guns for both studies. We also do not want to extend the time the fish are anesthetized while switching to another setup in-between fish (extended anesthetizing also compromises survival). The fish are placed into different release locations depending on the project.

To ensure that the effects of tagging are equal across studies (see Analysis), it is ideal that all fish are tagged with the same tagging equipment. If different systems are used, it would require that taggers have equivalent skill and experience with both. For example, different tagging guns will have different characteristics (e.g., bevel direction and trigger sensitivity) so users would need to be proficient with both guns to ensure equal effect on fish survival due to tag wound placement and insertion depth. This latter is a critical variable because if the tags are inserted too shallow, they pop out and are lost and if too deeply they can be inserted into internal organs, which can kill the fish.

These two projects tag tens of thousands of fish per season and so it is critical to keep a complicated process as simple as possible. Not only does the tagging process need to be accomplished quickly for a given fish, but also from fish-to-fish exacerbating the need for a streamlined process most successfully executed using identical equipment. BPA already provides NOAA with trays of Biomark tags for their half of the tagging.

- Analysis - A feature of this joint research not only involves the concurrent tagging, but also entails comparative statistical analyses using both sets of tagged fish. These tagged fish will be used to evaluate salmon survival through two different migration routes. Survival is modeled based on detections of the tagged fish in their downstream juvenile migration and on their adult return migration. The goal is to assess the difference in survival between the two groups. Therefore, it is very important for the tagged fish to be assumed completely equal on all survival factors other than their migration route. If they are sampled, processed, and tagged the same way, we can be confident in that assumption. If they are tagged with different systems that is not as easily assumed. If a different system makes a different needle wound, or places tags differently within the fish body, it could lead to different survival or even tag-retention rates and violate the assumption. These studies are designed to provide information to manage the success of threatened or endangered salmon, and thus it is mission critical that they be carried out to best of our abilities to provide that information.

5. Market Research.

We investigated several different options that helped us to determine that Biomark had the lowest price and would not require us to purchase new injector guns. Attached is a file containing the different quotes we received. In general, the other options had obstacles with doing the work on such a quantity of PIT-tags, especially in the needed time frame. Other vendors either wanted us to buy their tags or their set up of injector guns, assemblages and trays. At this time, we are unable to determine the performance of these setups; we would need to test the injector guns from these other vendors in terms of “trigger sensitivity” or other factors that could directly impact fish survival. This

effort does not seem worth it considering the benefits of maintaining our compatibility with the BPA-funded project that we rely on for our comparison analyses and that already supplies NOAA with loaded trays of tags from Biomark.